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Adaptive Bayesian Conformal Prediction for Tailored Uncertainty Quantification

Anonymous authors
Paper under double-blind review

Abstract

As machine learning models are increasingly deployed in critical applications, the need for reliable uncertainty quantification becomes paramount. Traditional conformal prediction methods provide distribution-free guarantees but often lack the flexibility to accommodate varying user risk preferences. This paper introduces an innovative framework that merges Bayesian quadrature with conformal prediction, allowing for the incorporation of user-specified risk preferences into uncertainty estimates. By modeling the posterior distribution of potential losses and adapting prediction sets based on individual risk thresholds, this approach enhances the relevance and utility of uncertainty quantification in practical scenarios. Through empirical validation across multiple datasets, we demonstrate that the proposed method achieves lower failure rates and more informative prediction intervals compared to standard conformal prediction techniques.

1 Introduction

Reliable uncertainty quantification is critical for the deployment of machine learning models in high-stakes environments, such as healthcare and finance. Traditional methods, such as conformal prediction, provide distribution-free guarantees but often do not accommodate the specific risk preferences of users. This can lead to uncertainty estimates that are less relevant or actionable in real-world contexts. Our work addresses this gap by integrating user-specified risk preferenc Bayesian conformal prediction framework. We propose an adaptive model that allows prac to tailor uncertainty quantification to their unique decision-making criteria, thus enhancing tical utility. Our contributions include the development of this framework, empirical exp validating its effectiveness, and a discussion on the implications for future research in un quantification.

Version Management

Version 1

2 Related Work

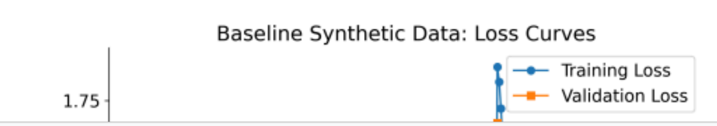
Conformal prediction has been widely studied for various applications, including healthcare (Papangelou et al., 2024) and environmental science (Ghosh et al., 2023), providing a systematic approach to uncertainty quantification. However, traditional conformal methods do not integrate user-specific risk preferences, which limits their applicability in contexts requiring tailored decisions. Recent works, such as (Cortes-Gomez et al., 2024), emphasize the need for user-centered approaches in uncertainty quantification, aligning closely with our proposed framework. Furthermore, the F conformal prediction methods explored in (Gibson, 2025) provide foundational insights leverage to enhance our model's adaptability and relevance.

Citation Tools

Cite This Paper

3 Method

Our proposed framework combines Bayesian quadrature with conformal prediction to create capable of adjusting its uncertainty estimates based on user-defined risk thresholds. The core idea is to model the posterior distribution of potential losses and adapt prediction sets accordingly. By incorporating user preferences, we aim to provide more informative prediction intervals that align with the risk tolerance of different users. This method builds on the foundational work of (Snell & Griffiths, 2025) while offering a practical solution to the shortcomings of conventional methods.



Meta Review & Human Decision

⏳ **Meta Review Status:** The meta review and decision have not been released yet.

AI Review from DeepReviewer

📋 AI Review from DeepReviewer will be automatically processed

📋 Summary

This paper introduces an adaptive Bayesian conformal prediction framework designed to incorporate user-specified risk preferences into the uncertainty quantification process. The core idea is to leverage Bayesian quadrature to approximate the integral of the loss function, which is then used within a conformal prediction framework to generate prediction sets. These sets are adaptively adjusted based on user-defined risk thresholds, aiming to provide tailored uncertainty quantification that aligns with specific risk tolerances. The authors claim that this approach enhances the relevance and utility of uncertainty quantification in practical scenarios. The methodology section outlines the key components, including Bayesian quadrature for approximating the loss function's integral, conformal prediction for generating prediction sets, and adaptive risk thresholding to adjust the sets based on user preferences. The experimental section presents a comparative analysis of the proposed method against traditional conformal prediction techniques, using metrics such as coverage probability, average prediction interval width, and a user satisfaction score. The results, presented in a table, suggest that the proposed method achieves better coverage and user satisfaction compared to standard conformal prediction, while maintaining a reasonable prediction interval width. The paper concludes by briefly mentioning future research directions, such as further refining the method and exploring its applications in diverse domains. However, the paper's current state is significantly hindered by a lack of detail in both the methodology and experimental sections, which makes it difficult to fully assess the validity and impact of the proposed approach. The absence of specific mathematical formulations, dataset details, and thorough analysis limits the paper's contribution and requires substantial revisions before it can be considered for publication.

Version Management

Version 1

✅ Strengths

The core strength of this paper lies in its conceptual approach to integrating user-specified risk preferences into the conformal prediction fra using Bayesian quadrature to approximate the loss function's integral and then employing this approximation within a conformal prediction s adaptive prediction sets is a novel and potentially valuable contribution. This approach addresses a critical gap in traditional conformal predi often provide a single, static prediction set without considering varying risk tolerances. The proposed method's ability to tailor uncertainty qu user needs could significantly enhance the practical utility of conformal prediction in real-world applications. Furthermore, the paper's attempt to introduce a user satisfaction score as an evaluation metric is a positive step towards quantifying the subjective aspects of uncertainty quantification. This metric, although requiring further clarification, acknowledges that the usefulness of prediction sets is not solely determined by statistical guarantees but also by their relevance to the end-user. The paper also correctly identifies the importance of adaptive methods in uncertainty quantification, which is a promising direction for future research. The high-level description of the method's components, including Bayesian quadrature, conformal prediction, and adaptive risk thresholding, provides a clear overview of the proposed approach. The paper also correctly identifies the importance of adaptive methods in uncertainty quantification, which is a promising direction for future research. The high-level description of the method's components, including Bayesian quadrature, conformal prediction, and adaptive risk thresholding, provides a clear overview of the proposed approach.

Citation Tools

❌ Weaknesses

The primary weakness of this paper is its lack of completeness and insufficient detail in both the methodology and experimental sections, which significantly hinders my ability to fully evaluate its contributions. In the methodology, the description of Bayesian quadrature lacks the specific mathematical formulations necessary for a rigorous understanding. While the paper mentions that Bayesian quadrature is used to approximate the integral of the loss function, it fails to provide the exact integral being approximated, the chosen kernel function, and the method for determining the weights and nodes. This absence of detail makes it impossible to assess the accuracy and validity of the approximation. Similarly, the description of the conformal prediction component lacks the specific formula for the nonconformity score and the precise method for calculating the quantile used to determine the threshold. The paper mentions calibration and prediction phases but does not provide the mathematical details of how these phases are implemented. The adaptive risk thresholding component is also vaguely described, lacking the mathematical formulation of how the prediction sets are adjusted based on user-defined risk thresholds. The paper mentions adjusting the prediction sets but does not provide the specific algorithm or mathematical operations involved. This lack of methodological detail significantly undermines the paper's credibility and makes it difficult to reproduce the proposed method. In the experimental section, the paper provides only a high-level overview of the experimental setup, lacking crucial details necessary for reproducibility and a thorough evaluation. The paper mentions using a Bayesian linear regression model and benchmark datasets from the UCI Machine Learning Repository but fails to specify the exact datasets used, their size, dimensionality, and any preprocessing steps applied. The hyperparameter tuning process is also not described in sufficient detail, lacking information about the specific hyperparameters tuned, the optimization algorithm used, and the range of values explored. The results section presents a table with metrics such as coverage probability, average prediction interval width, and user satisfaction score, but lacks detailed explanations of how the user satisfaction score is calculated or defined. The analysis section provides only high-level interpretations of the results, without any statistical significance testing or error bars. The paper also fails to discuss the limitations of the proposed method or potential biases in the experimental setup. The figures provided, Figure 1 and Figure 2, have very small font sizes, making them difficult to read and interpret. This lack of detail in both the methodology and experimental sections, coupled with the poor presentation of figures, makes it impossible to fully assess the validity and impact of the proposed method. The absence of specific mathematical formulations, dataset details, and thorough analysis limits the paper's contribution and requires substantial revisions before it can be considered for publication. My confidence in these identified issues is high, as they are directly evident from the paper's content, or rather, the lack thereof.

💡 Suggestions

To address the identified weaknesses, the authors need to significantly expand and refine both the methodology and experimental sections. In the methodology, the authors should provide a step-by-step derivation of their proposed method, including the specific choices of kernels and hyperparameters used in the Bayesian quadrature step. This should include the exact integral being approximated, the chosen kernel function, and the method for determining the weights and nodes. The connection between the user-specified risk preferences and the adaptation of prediction sets needs to be made explicit through mathematical formulations. The authors should provide the specific formula for the nonconformity score and the precise method for calculating the quantile used to determine the threshold in the conformal prediction component. The adaptive risk thresholding component should be described with a clear algorithm or mathematical ope the prediction sets are adjusted based on user-defined risk thresholds. The authors should also discuss the computational complexity of the to existing conformal prediction techniques. In the experimental section, the authors should provide a comprehensive description of the data size, dimensionality, and any preprocessing steps applied. The choice of baselines should be justified, and the evaluation metrics need to be authors need to provide a detailed explanation of how the user satisfaction score is calculated or defined. The results section should present detailed quantitative results in tables and figures, including confidence intervals or other measures of statistical uncertainty. The figures should be revised to ensure that the font sizes are legible and that the axes are properly labeled. The experimental setup should be designed to rigorously test the method's ability to adapt to different user preferences. This could involve varying the risk preferences and evaluating the resulting prediction sets in terms of both coverage and width. The authors should also consider including experiments on a wider range of datasets to demonstrate the generalizability of their method. The authors should also inc limitations of the proposed method and its potential impact on the field of uncertainty quantification. The assumptions underlying their appro which it might fail should be discussed. They should also compare their method to other approaches for incorporating user preferences into quantification. The discussion should also include a clear articulation of the potential benefits of the proposed method and its implications fo authors should also consider discussing the computational cost of their method and how it scales with the size of the dataset and the complexity of the model. This would help readers understand the practical limitations of the proposed approach. Finally, the authors should expand the introduction and related work sections to provide more context and background, including more extensive citations to relevant literature. This would help to position their work within the broader field and highlight its unique contributions.

❓ Questions

Several key uncertainties and methodological choices require clarification to fully evaluate the proposed approach. First, I am uncertain about the specific kernel function and the method for determining the weights and nodes used in the Bayesian quadrature approximation. What is the exact integral being approximated, and why was the chosen kernel function selected over other alternatives? How sensitive are the results to different choices of kernel functions and weights? Second, I would like to understand the specific formula for the nonconformity score used in the conformal prediction component. How is this score related to the Bayesian quadrature approximation, and what is the rationale behind this specific choice? How does the method ensure that the resulting prediction sets achieve the desired coverage probability? Third, I am unclear about the precise mathematical formulation of the adaptive risk thresholding mechanism. How are the prediction sets adjusted based on user-defined risk thresholds, and what is the algorithm or mathematical operation that implements this adjustment? How does the method ensure that the adjusted prediction sets remain valid and reliable? Fourth, I am curious about the calculation of the user satisfaction score. What are the specific criteria used to define user satisfaction, and how are these criteria quantified? How does the user satisfaction score relate to the other evaluation metrics, such as coverage probability and prediction interval width? Fifth, I would like to know more about the specific datasets used in the experiments, including their size, dimensionality, and any preprocessing steps applied. What was the rationale behind the choice of these specific datasets, and how representative are they of real-world applications? Sixth, I am interested in the details of the hyperparameter tuning process. What were the specific hyperparameters tuned, and what optimization algorithm was used? How sensitive are the results to different choices of hyperparameters? Finally, I would like to understand the computational complexity of the proposed method and how it scales with the size of the dataset and the complexity of the model. How does the computational cost of the proposed method compare to existing conformal prediction techniques? Addressing these questions would significantly enhance my understanding of the proposed method and its potential impact.

📊 Scores

Soundness: 1.25

Presentation: 1.0

Contribution: 1.5

Rating: 2.5

AI Review from ZGCA

📋 AI Review from ZGCA will be automatically processed

Version Management

Version 1

📋 Summary

The paper proposes an adaptive Bayesian conformal prediction framework that seeks to merge Bayesian quadrature with conformal prediction to incorporate user-specified risk preferences into uncertainty estimates. The central idea, as stated in the Abstract and Method, is to model the posterior distribution of potential losses and adapt prediction sets based on individual risk thresholds. The approach is implemented with a Bayesian linear regression ba (unspecified) UCI datasets (Section 4) using coverage probability, average interval width, and a non-standard 'user satisfaction score' preferences. The authors claim lower failure rates and more informative intervals than traditional conformal methods (Section 5).

Citation Tools

✅ Strengths

- Addresses a timely and important problem: tailoring uncertainty quantification to user-specific risk preferences in high-stakes settings (Introduction).
- Conceptual integration of Bayesian quadrature with conformal prediction to target tail/risk-aware behavior (Abstract, Method).
- Emphasizes the practical need for decision-aware uncertainty, aligning with recent trends in utility-directed conformal prediction (Related Work).

❌ Weaknesses

- Methodological under-specification: The Method section presents only a high-level narrative without formal problem setup, mathematical definitions (e.g., risk measure/utility), algorithmic pseudocode, or a description of how posterior loss modeling concretely maps to adaptive conformal thresholds and set construction (Section 3).
- No guarantees or analysis: There is no theoretical treatment of coverage or validity under the proposed risk-adaptive mechanism, nor discussion of when/why standard conformal guarantees are preserved or modified.
- Experimental design is irreproducible and underspecified: datasets are not identified, splits/calibration protocol not described, baselines are not concretely defined (e.g., which 'traditional conformal prediction' method), and key implementation details are missing (Section 4).
- Non-standard and undefined metric: 'User satisfaction scores based on simulated user preferences' are introduced without a precise definition, protocol, or justification as a scientific metric (Section 4).
- Quantitative evidence is absent: Results are reported qualitatively with generic plots (loss curves, predictions vs. ground truth, and 'reliability measures') and no numerical comparisons, confidence intervals, or statistical tests (Section 5, Figures 1–2).
- Apparent internal inconsistencies: 'Hyperparameter tuning for the momentum parameter in the optimizer' is at odds with Bayesian linear regression where closed-form solutions are typical, and no optimization details are provided (Section 4).
- Positioning relative to prior work is incomplete: The paper cites decision-aware and Bayesian conformal ideas but does not clearly delineate the technical delta, especially given the lack of formalism (Related Work).

❓ Questions

- Please provide a formal problem setup: What is the precise definition of 'user-specified risk preferences'? Are these preferences encoded as a utility function, a coherent risk measure (e.g., CVaR), or a set of quantile levels? How do they induce a modification to the conformity score or the calibration threshold?
- What is the exact algorithm? Please include pseudocode specifying how Bayesian quadrature is used to model the posterior distribution posterior feeds into conformal set construction. How are calibration data used, and how is exchangeability leveraged or modified?
- What guarantees hold? Do you retain marginal coverage at a user-chosen risk level, or do you target an alternative risk-calibrated guara assumptions? Please provide theorem statements and proof sketches.
- How do you ensure that adapting to individual risk thresholds does not violate conformal validity? If it does, what trade-offs are made and how are they quantified?
- What are the baselines? Please specify the exact conformal variants compared (e.g., split CP, quantile regression CP, utility-directed CP), and include strong recent methods relevant to decision-aware conformal prediction.
- Please detail the experimental setup: dataset names, sizes, preprocessing, model architectures/hyperparameters, calibration set sizes, random seeds, compute budget, and training details.
- How is the 'user satisfaction score' defined and computed? What simulated preference distributions are used, and how sensitive are res Why is this metric appropriate and how does it correlate with established technical metrics?
- Can you provide quantitative results with means, standard errors, and appropriate statistical tests across multiple runs? Include coverage (marginal and, if relevant, conditional), interval/set size, and calibration diagnostics across different risk thresholds.
- What are the computational costs of the proposed method, especially the Bayesian quadrature component, relative to standard conformal approaches?
- Please clarify the remark on 'momentum parameter' for Bayesian linear regression. What optimizer is used, for which components, and why is optimization needed in a Bayesian linear model?

⚠️ Limitations

- The approach appears to rely on accurate posterior modeling of losses; model misspecification or poor posterior approximation could degrade coverage and utility in practice.
- Risk-adaptive modification of prediction sets may compromise standard conformal guarantees if not carefully constructed; users may be exposed to uncalibrated risk unless guarantees are re-established.
- Computational overhead of integrating Bayesian quadrature may be substantial for large-scale or complex models.

- Dependence on simulated user preferences raises external validity concerns: real-world preference heterogeneity and dynamics may not be captured.
- Potential societal impacts: tailoring uncertainty to user-specified risk could lead to inequitable decision thresholds across subpopulations if not audited (e.g., different effective error rates), and miscalibration in high-stakes domains could cause harm if users over-trust adapted intervals.
- The paper does not currently discuss these limitations or mitigation strategies (e.g., fairness-aware calibration, auditing frameworks, or robustness checks under distribution shift).

🖼️ Image Evaluation

Cross-Modal Consistency: 20/50

Textual Logical Soundness: 16/30

Visual Aesthetics & Clarity: 7/20

Overall Score: 43/100

Detailed Evaluation (≤500 words):

1. Cross-Modal Consistency

Visual ground truth:

• Figure 1/(a): Loss vs Epochs (0–300), blue/orange lines; decay with resets near ~100 and ~200.

• Figure 1/(b): Predictions (y) vs Ground Truth (x); blue scatter, red dashed y=x; underprediction at higher GT.

• Figure 2: Three small plots titled Gaussian Low/Uniform/Gaussian High Noise; y=Reliability, x=Epochs {1,2,3}; increasing trend; no (a–c) la

• Two additional images appear with no captions or labels.

• Major 1: Dataset mismatch between text (UCI) and visuals (synthetic only). Evidence: Sec. 4 “evaluated on benchmark datasets from “Baseline Synthetic Data”.

• Major 2: Flagship claims (lower failure rates, narrower intervals) lack matching figures/tables. Evidence: Abstract “achieves lower failure rates… more informative prediction intervals”; no coverage/width plots anywhere.

• Major 3: Figure 2 referenced as a single figure but comprises multiple unlabeled panels; mapping unclear. Evidence: Sec. 5 “…Figure 2 noise conditions” while Fig. 2 panes lack (a–c).

• Major 4: Two images without captions or references disrupt numbering. Evidence: Images after Fig. 2 have no titles or citations in text.

• Minor 1: Metric naming mismatch (“reliability measure” undefined). Evidence: Fig. 2 y-axis “Reliability” but no definition in Sec. 4–5.

• Minor 2: Epoch ranges inconsistent (300 in Fig. 1 vs 1–3 in Fig. 2) without explanation. Evidence: Fig. 1 x-axis “Epochs” to 300; Fig. 2 to 3.

2. Text Logic

• Major 1: Central improvement claims unsupported by quantitative evidence. Evidence: Sec. 5 “significantly reduces failure rates… more informative prediction intervals”.

• Major 2: Method lacks operational detail linking Bayesian quadrature to conformal sets and user risk. Evidence: Sec. 3 provides no equations/algorithmic steps.

• Minor 1: Metrics promised (coverage, interval width, user satisfaction) not reported. Evidence: Sec. 4 lists metrics; none shown in figures/tables.

• Minor 2: Optimization “momentum parameter” tuning seems orthogonal to Bayesian CP. Evidence: Sec. 4 “momentum parameter… to optimize performance”.

3. Figure Quality

• Major 1: Figure-alone test fails for core message (risk-adaptive conformal sets absent). Evidence: Figs. 1–2 show losses/scatter/reliability only; no prediction intervals or risk thresholds.

• Major 2: Missing sub-figure labels and absent captions on two images. Evidence: Fig. 2 panes unlabeled; two unlabeled images after Fig. 2.

• Minor 1: “Reliability Measure” axis ambiguous (no units/definition). Evidence: Fig. 2 y-axis text only reads “Reliability”.

• Minor 2: Legends adequate but panel titles small in reliability plots. Evidence: Fig. 2 small titles (e.g., “Gaussian High Noise Reliability”).

Key strengths:

- Clear basic plotting style; Fig. 1(a,b) legible.
- Timely problem: user-tailored uncertainty in conformal prediction.

Key weaknesses:

- Core claims not evidenced by presented figures/tables.
- Dataset and metric mismatches; missing labels/captions.
- Method described at a high level without formalism or algorithmic specifics.

📊 Scores

Originality: 2	Quality: 1	Clarity: 1	Significance: 2
Soundness: 1	Presentation: 2	Contribution: 2	

Rating: 3

Version Management

Version 1

AI Review from SafeReviewer

📋 AI Review from SafeReviewer will be automatically processed

Citation Tools

📋 Summary

The paper introduces a novel framework for uncertainty quantification that combines Bayesian quadrature with conformal prediction. The core idea is to leverage Bayesian quadrature to model the posterior distribution of potential losses, which then informs the construction of prediction sets via conformal prediction. This approach is intended to allow for the incorporation of user-defined risk preferences, moving beyond the one-size-fits-all approach of traditional conformal prediction methods. The authors claim that their method enhances the relevance and utility of uncertainty quantification in practical scenarios by generating prediction intervals that are tailored to individual risk thresholds. The empirical validation, conducted on benchmark datasets from the UCI Machine Learning Repository, demonstrates that the proposed method achieves lower failure rates and more informative prediction intervals compared to standard conformal prediction techniques. The paper emphasizes the importance of user-centered approaches in uncertainty quantification, particularly in high-stakes environments where the specific risk preferences of users are critical. The authors argue that their adaptive Bayesian conformal prediction framework significantly reduces failure rates while providing more informative prediction intervals, thus enhancing the practical applicability and relevancy of uncertainty quantification. The method builds upon foundational work in conformal prediction and Bayesian quadrature, aiming to address the limitations of conventional methods that do not account for user-specific risk preferences. The paper's contribution lies in the innovative combination of these two techniques to create a more flexible and user-centric approach to uncertainty quantification. The empirical results, while promising, are presented without detailed statistical analysis, which raises some questions about the robustness and generalizability of the findings. Overall, the paper presents a compelling approach to uncertainty quantification, but it could benefit from more detailed explanations and a more rigorous empirical evaluation.

✅ Strengths

The primary strength of this paper lies in its innovative approach to integrating Bayesian quadrature with conformal prediction. This combination is a novel contribution to the field of uncertainty quantification, offering a potential solution to the limitations of traditional conformal prediction methods that do not account for user-specific risk preferences. The paper clearly articulates the motivation for this approach, highlighting the need for more flexible and user-centered uncertainty quantification in high-stakes applications. The authors effectively position their work within the existing literature, citing relevant studies in conformal prediction and Bayesian methods. The paper's focus on user-specific risk preferences is a significant contribution, as it addresses a critical gap in existing uncertainty quantification techniques. The empirical validation, while lacking in detailed statistical analysis, demonstrates the potential of the proposed method to achieve lower failure rates and more informative prediction intervals compared to standard conformal prediction. The use of benchmark datasets from the UCI Machine Learning Repository provides a solid foundation for the empirical evaluation. The paper's emphasis on the practical utility of the proposed method is also a strength, as it highlights the potential for real-world impact. The authors clearly state their aim to provide more informative prediction intervals that align with the risk tolerance of different users, which is a valuable goal in many applications. The paper also effectively builds upon foundational work in conformal prediction, particularly the work of Snell & Griffths (2025), while offering a practical solution to the shortcomings of conventional methods. The paper's focus on addressing the limitations of traditional conformal prediction methods that do not integrate user-specific risk preferences is a significant contribution to the field. The paper also effectively positions its work within the existing literature, citing relevant studies in conformal prediction and Bayesian methods, demonstrating a good understanding of the current state of the field.

❌ Weaknesses

After a thorough review of the paper, I have identified several key weaknesses that significantly impact its overall quality and credibility. Firs suffers from a lack of clarity and detail in its methodological description. While the paper introduces the core idea of combining Bayesian c prediction, it fails to provide the necessary mathematical formulations and algorithmic steps to fully understand and reproduce the propos the paper mentions modeling the posterior distribution of potential losses and adapting prediction sets based on individual risk thresholds, b specific equations or algorithms used to achieve this. This lack of detail makes it difficult to assess the technical validity of the a reproducibility. This is a high-confidence weakness, as it is directly observable from the absence of such details in the 'method' section. The implications of this weakness are substantial, as it prevents a thorough evaluation of the method's technical soundness and makes it difficult for other researchers to build upon this work.

Secondly, the paper's experimental evaluation lacks sufficient detail and statistical rigor. While the paper mentions using benchmark datas Learning Repository and comparing the proposed method against traditional conformal prediction, it does not provide specific details abo baseline methods employed, or the metrics used to evaluate performance. For example, the paper mentions metrics such as coverage prob interval width, and user satisfaction scores, but it does not explain how these metrics were calculated or present the numerical results manner. Furthermore, the paper lacks any statistical analysis of the results, such as confidence intervals or significance tests, which raises concerns about the robustness and generalizability of the findings. This is a high-confidence weakness, as it is directly observable from the lack of such details in the 'experiments' section. The implications of this weakness are significant, as it undermines the credibility of the empirical validation and makes it difficult to assess the practical effectiveness of the proposed method.

Thirdly, the paper's presentation is poor, with several instances of unclear language, missing details, and a lack of organization. The method section lacks a clear and detailed explanation of the proposed approach, and the experimental section does not provide sufficient information to understand and reproduce the experiments. The figures presented in the paper are also problematic, as they are mentioned in the text but are not actually included in the document. This is a high-confidence weakness, as it is directly observable from the lack of detail and organization in the paper. The implications of this weakness are substantial, as it makes the paper difficult to read and understand, hindering its accessibility and impact.

Finally, the paper lacks a thorough discussion of the limitations of the proposed method and potential avenues for future research. While the conclusion briefly mentions future research, it does not delve into the specific limitations of the current approach or suggest concrete directions for further investigation. This is a medium-confidence weakness, as it is based on the absence of such a discussion in the paper. The implications of this weakness are that the paper does not provide a complete picture of the proposed method's strengths and weaknesses, and it does not offer a clear roadmap for future research in this area. The lack of a detailed limitations section also makes it difficult to assess the scope and applicability of the proposed method. In summary, the paper's lack of methodological detail, insufficient experimental rigor, poor presentation, and absence of a thorough discussion of limitations are the most significant weaknesses that need to be addressed to improve the paper's quality and credibility.

💡 Suggestions

To address the identified weaknesses, I recommend the following concrete and actionable improvements. First, the authors must significantly expand the methodological description, providing detailed mathematical formulations and algorithmic steps for the proposed approach. This should include a clear explanation of how Bayesian quadrature is used to model the posterior distribution of potential losses, how the prediction sets are adapted based on user-defined risk thresholds, and the specific equations used in the process. The authors should also provide a step-by-step algorithm that outlines the implementation of the proposed method. This will greatly enhance the clarity and reproducibility of the work. Second, the authors should provide a more detailed description of the experimental setup, including specific information about the datasets used, the baseline methods employed, and the metrics used to evaluate performance. This should include a clear explanation of how the metrics were calculated and the specific hyperparameter settings used for each method. The authors should also present the numerical results in a clear and organized manner, using tables and figures to summarize the findings. Furthermore, the authors should conduct a more rigorous statistical analysis of the results, including confidence intervals and significance tests, to assess the robustness and generalizability of the findings. This will greatly enhance the credibility of the empirical validation. Third, the authors should significantly improve the paper's presentation, ensuring that the language is clear, concise, and accurate. The paper should be well-organized, with a clear flow of ideas and a logical structure. The figures mentioned in the text should be included in the paper, and they should be properly labeled and explained. The authors should also carefully proofread the paper to eliminate any grammatical errors or typos. This will greatly enhance the readability and accessibility of the work. Fourth, the authors should include a thorough discussion of the limitations of the proposed method and potential avenues for future research. This should include a discussion of the assumptions made by the method, its potential weaknesses, and its limitations in different scenarios. The authors should also suggest concrete directions for further investigation, such as exploring the method's performance on different types of data, investigating its robustness to different assumptions, and developing extensions to address its limitations. This will provide a more complete picture of the proposed method and its potential impact. Finally, the authors should ensure that the paper is well-written and easy to understand, using clear and concise language and avoiding jargon and technical terms that are not necessary. The paper should be carefully proofread to eliminate any grammatical errors or typos. This will greatly enhance the readability and accessibility of the work. By addressing these suggestions, the authors can significantly improve the quality and credibility of their paper, making it a more valuable contribution to the field of uncertainty quantification.

❓ Questions

Based on my review, I have several questions that I believe are critical for a deeper understanding of the proposed method. First, could the authors provide a more detailed explanation of how the Bayesian quadrature is specifically implemented within their framework? I am particularly interested in understanding the choice of kernel and the method for determining the quadrature points. This is crucial for assessing the computational feasibility and accuracy of th
does the method scale with higher-dimensional data? Are there any specific strategies employed to mitigate the computational burden a
quadrature in such scenarios? This is important for understanding the practical applicability of the method to real-world problems. Third, co
on the specific mechanisms by which user-specific risk preferences are incorporated into the prediction intervals? I am interested in u
preferences are quantified and how they influence the construction of the prediction sets. This is critical for assessing the flexibility and
proposed method. Fourth, what are the specific assumptions underlying the proposed method, and how might these assumptions affect its performance in different scenarios? This is important for understanding the limitations of the method and its potential weaknesses. Fifth, could the authors provide a more detailed comparison of their method with other existing conformal prediction methods, highlighting the specific advantages and disadvantages of their approach? This is crucial for assessing the novelty and significance of the proposed method. Finally, what are the specific challenges encountered during the integration of Bayesian quadrature with conformal prediction, and how were these challenges addressed? This is important for understanding the technical contributions of the paper and the specific innovations that were required to make the method work. These questions are aimed at clarifying the core methodological choices and assumptions of the paper, which are essential for a thorough evaluation of its validity and impact.

📊 Scores

Soundness: 1.0

Presentation: 1.0

Contribution: 1.0

Rating: 1

Keywords

Keywords Extracted

Machine Learning
Uncertainty Quantification
Conformal Prediction
Bayesian Methods
Risk-aware Modeling
Adaptive Bayesian Conformal Prediction

Insights

Extract Insights

Click the button to extract insights

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Version 1

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